

# CYBERPRO AI INTERNSHIP

## CyberPro AI Assessment Report

CPAI-2026-001

Breaking Barriers Assessment

Part I - Reconnaissance and Vulnerability Analysis

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Course: CPIN-280 Cyber Security Internship

Platform: Cympire Cyber Range

Assessment Date: June 2026

Assessment Status: IN PROGRESS

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## Bottom Line Up Front (BLUF)

The **Breaking Barriers** assessment successfully completed the reconnaissance and vulnerability identification phases, resulting in the identification of **vsftpd 2.3.4** and its associated critical vulnerability, **CVE-2011-2523**. Progress into the exploitation phase was interrupted by a Metasploit Framework initialization error within the provided Kali Linux environment, preventing live validation of the identified vulnerability. All completed assessment activities were documented, and the remaining exploitation and post-exploitation objectives will be completed once the environment issue has been resolved.

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# 1. Assessment Background

The **Breaking Barriers** exercise is a guided penetration testing assessment within the Cympire Cyber Range. Rather than functioning as a traditional Capture-the-Flag exercise, the assessment emphasizes following a structured penetration testing methodology while documenting observations throughout the engagement. Students are expected to think like security consultants by identifying vulnerabilities, supporting technical findings with evidence, and communicating results in a manner understandable to both technical and non-technical audiences.

The assessment officially began with the selection of the Breaking Barriers exercise within the Cympire Cyber Range environment. This established the scope of the engagement and provided the operational environment used throughout the assessment. The initial exercise selection is documented in **Figure 1**, establishing the starting point for the engagement.

Before any technical activity was performed, the exercise briefing was reviewed to understand the mission objectives, expected workflow, and assessment scenario. Reviewing the briefing ensured that subsequent technical activities remained aligned with the intended methodology rather than approaching the exercise as a simple exploitation challenge. The original mission briefing is preserved as **Figure 2** within Appendix A.

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## 2. Assessment Objectives

The objective of this assessment was to conduct a structured security assessment by following an industry-standard penetration testing workflow. Primary objectives included identifying exposed network services, determining software versions, researching known vulnerabilities, selecting appropriate exploitation techniques, and documenting all findings throughout the engagement.

Equally important was the development of professional reporting habits. Throughout the Week 3 CyberProAI meeting, emphasis was placed on documenting work continuously instead of attempting to reconstruct the assessment afterward. This methodology was followed throughout the exercise by capturing evidence immediately after each major milestone.

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## 3. Assessment Methodology

The assessment began with network reconnaissance using **Nmap** to identify accessible services exposed by the target system. Service enumeration successfully identified multiple network services and provided software version information necessary for subsequent vulnerability research.

Enumeration confirmed that the target FTP service was running **vsftpd 2.3.4**, establishing the basis for additional investigation. The original Nmap output documenting the service fingerprint is presented in **Figure 3**, providing the primary evidence supporting the version identification.

Once the software version had been verified, publicly available vulnerability intelligence was consulted to determine whether the identified version contained any known security weaknesses. Research confirmed that **vsftpd 2.3.4** corresponds to **CVE-2011-2523**, a well-known backdoor vulnerability affecting compromised distributions of the software. The supporting vulnerability research is documented in **Figure 4**, illustrating the correlation between the enumerated service and the identified CVE.

Following completion of the reconnaissance and research phases, the assessment transitioned toward exploitation using the Metasploit Framework. Rather than reaching the expected Metasploit console, initialization consistently terminated with a permissions-related error within the provided Kali Linux environment. The observed error is preserved in **Figure 5**, documenting the operational condition that prevented progression into the exploitation phase.

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## 4. Technical Findings

Reconnaissance successfully identified the network services exposed by the target host and established an accurate software inventory for further analysis.

The most significant finding was the identification of an FTP service running **vsftpd 2.3.4**. Correlation with publicly available vulnerability intelligence identified **CVE-2011-2523**, a critical vulnerability historically associated with compromised distributions of that software version.

The completed reconnaissance and vulnerability analysis established a technically supported basis for exploitation. Although live validation could not be performed during this assessment session, sufficient evidence was collected to justify continued investigation once the required exploitation framework becomes operational.

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## 5. Challenges

The primary challenge encountered during this assessment occurred during the transition from vulnerability analysis into exploitation.

Repeated attempts to initialize the Metasploit Framework resulted in a permissions-related startup error within the provided Kali Linux environment. Basic troubleshooting confirmed that the Metasploit executable was present and that the initialization error could be reproduced consistently across multiple launch attempts.

Rather than attempting to bypass the issue or make unsupported conclusions, the assessment concluded with all completed work fully documented. The exploitation and post-exploitation phases remain pending until the environment issue can be resolved.

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## 6. Lessons Learned

This assessment reinforced that successful penetration testing depends upon disciplined methodology rather than simply executing exploits. Accurate reconnaissance, careful service enumeration, vulnerability research, and continuous documentation collectively establish the foundation for effective security assessments.

The exercise also demonstrated the practical value of vulnerability intelligence. Correctly identifying software versions and correlating them with known CVEs allows security professionals to prioritize investigation while understanding the potential business impact of identified weaknesses.

Finally, the assessment reinforced the importance of professional documentation. Capturing evidence immediately after each completed activity produced a complete chronological record of the engagement and eliminated the need to reconstruct findings after the assessment concluded.

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# Conclusion

Part I of the Breaking Barriers assessment successfully completed the reconnaissance and vulnerability analysis phases. The target system was enumerated, a critical vulnerability was identified through supporting research, and the assessment methodology was documented using professional reporting practices.

Progress into exploitation was prevented by a Metasploit initialization issue within the provided Kali Linux environment. Once the environment issue has been resolved, the remaining exploitation and post-exploitation activities will be completed and documented in **CyberPro AI Assessment Report CPAI-2026-002**, continuing the assessment from its current state.

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# Assessment Status

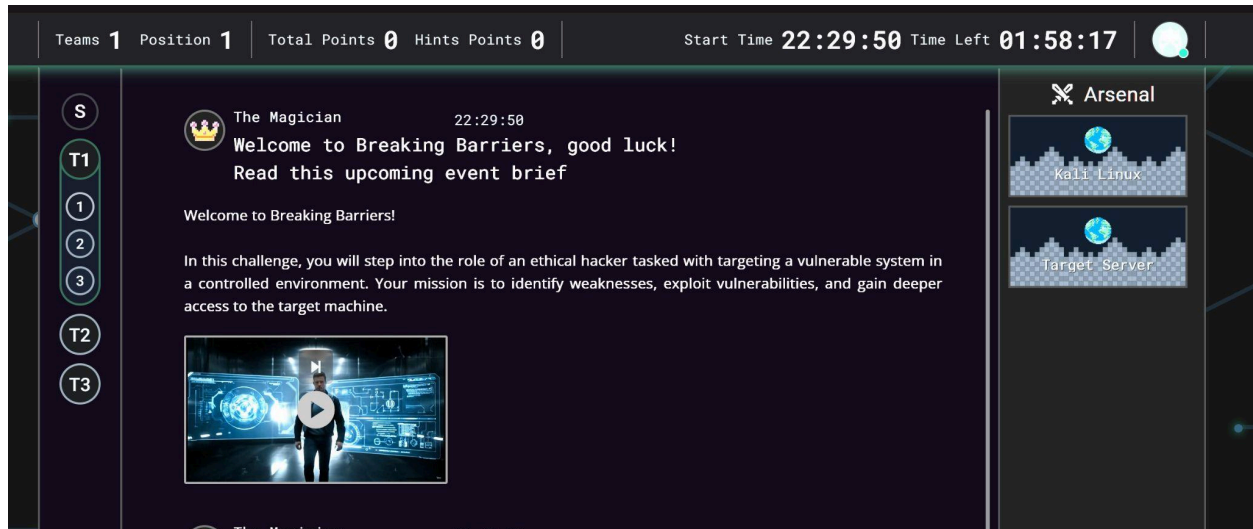
| Assessment Phase       | Status   |
|------------------------|----------|
| Reconnaissance         | Complete |
| Service Enumeration    | Complete |
| Vulnerability Analysis | Complete |
| Exploitation           | Pending  |
| Post-Exploitation      | Pending  |

**Overall Assessment State: IN PROGRESS**

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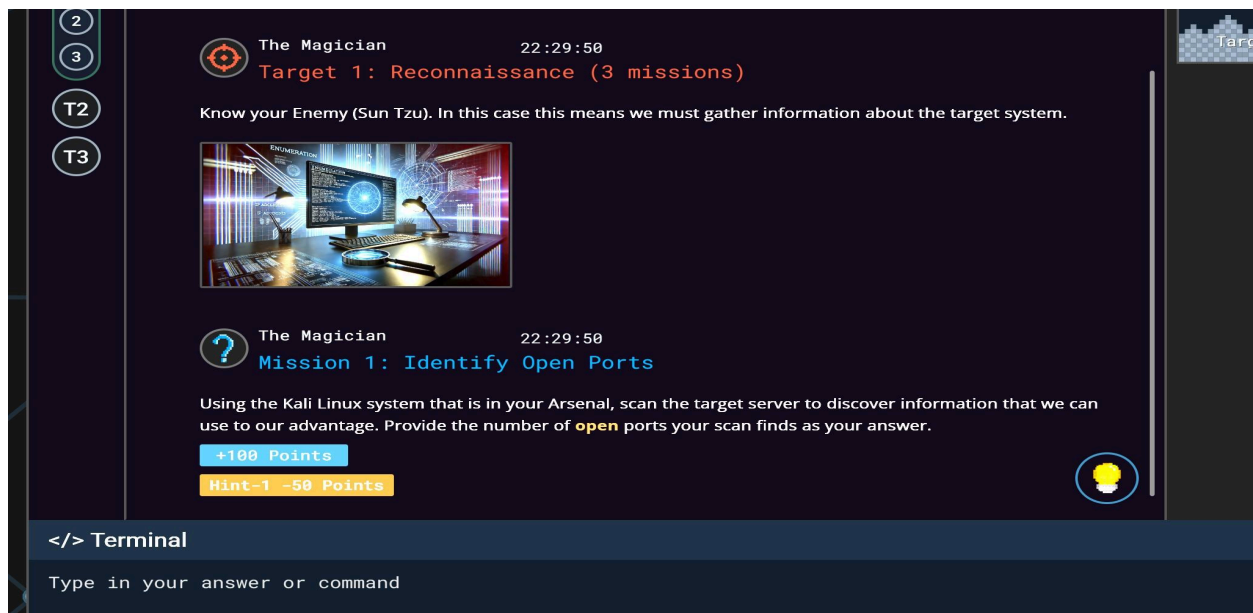
# Appendix A – Assessment Evidence

Figure 1. Breaking Barriers Exercise Selection



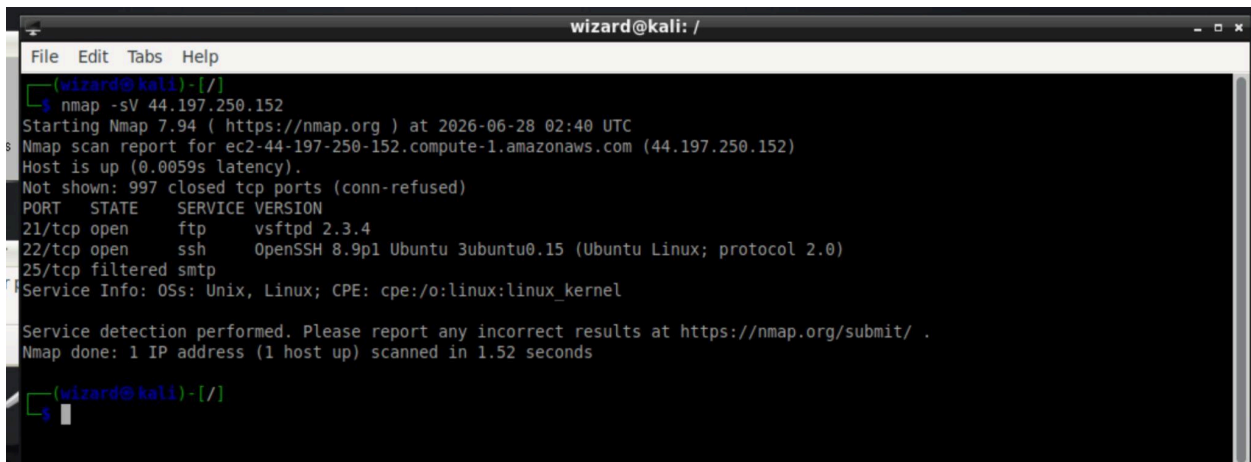
**Caption:** Exercise selection screen within the Cympire Cyber Range establishing the assessment environment.

Figure 2. Mission Briefing



**Caption:** Exercise briefing outlining the objectives, scenario, and expected penetration testing workflow.

Figure 3. Service Enumeration Results



Caption: Nmap reconnaissance identifying exposed network services, including the FTP service running **vsftpd 2.3.4**, which became the focus of the vulnerability assessment.

Figure 4. Vulnerability Research

VSFTPD 2.3.4 Backdoor Command Execution

TRY SURFACE COMMAND

← BACK TO SEARCH

| Disclosed   | Created      |
|-------------|--------------|
| Jul 3, 2011 | May 30, 2018 |

Description

This module exploits a malicious backdoor that was added to the VSFTPD download archive. This backdoor was introduced into the vsftpd-2.3.4.tar.gz archive between June 30th 2011 and July 1st 2011 according to the most recent information available. This backdoor was removed on July 3rd 2011.

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Caption: Research correlating the enumerated FTP service with **CVE-2011-2523**, establishing the technical basis for the planned exploitation phase.

**Figure 5. Metasploit Initialization Failure**

```
(wizard@kali)-[//]
└─$ msfconsole
<internal:dir>:98:in 'open': Permission denied @ dir_initialize - /usr/lib/ssl/private (Errno::EACCES)
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:50:in 'foreach'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:50:in 'walk'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:60:in 'block in wal
k'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:50:in 'foreach'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:50:in 'walk'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:60:in 'block in wal
k'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:50:in 'foreach'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:50:in 'walk'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path_scanner.rb:38:in 'call'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path.rb:93:in 'scan!'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/path.rb:77:in 'entries and dirs'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/cache.rb:164:in 'block (2 levels) i
n unshift_paths_locked'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/cache.rb:157:in 'reverse each'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/cache.rb:157:in 'block in unshift_p
aths_locked'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/store.rb:53:in 'block in transactio
n'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/store.rb:52:in 'synchronize'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/store.rb:52:in 'transaction'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/cache.rb:156:in 'unshift_paths_lock
ed'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/cache.rb:105:in 'block in unshift_p
aths'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/cache.rb:105:in 'synchronize'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/cache.rb:105:in 'unshift_paths'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/bootsnap-1.16.0/lib/bootsnap/load_path_cache/change_observer.rb:22:in 'unshift'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/railties-7.0.5/lib/rails/application.rb:342:in 'add lib to load_path!'
  from /usr/share/metasploit-framework/vendor/bundle/ruby/3.1.0/gems/railties-7.0.5/lib/rails/application.rb:73:in 'inherited'
```

**Caption:** Metasploit Framework initialization error encountered during the transition into exploitation, preventing live validation of the identified vulnerability and concluding Part I of the assessment.